

Swing Electrical Power Generator



Prepared by:
Riana Moodley
202313269

Department of Engineering
University of Zululand

Prepared for:
Mr J Mushid

Department of Engineering
University of Zululand

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Declaration



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Abstract

This project presents the design, simulation and the implementation of a swing electrical power generator which converts kinetic energy produced by a back-and-forth motion of a swing to usable electrical energy. The aim of this project is to find different methods of small-scale renewable energy generation using kinetic energy in an everyday occurrence. The system makes use of a rotational conversion component and an electrical generator to convert kinetic oscillatory motion into an electrical output. Simulation tools will be used to model the systems performance, followed by a practical implementation and testing to evaluate the efficiency of the system. This project highlights the potential of harvesting renewable energy into contributing to low-cost and low-power applications.

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1. Introduction

1.1 Background to the study

Energy is the ability to do work. It can be converted from one form to another such as heat, light, motion, chemical and electrical. In recent years, energy consumption has increased vastly due to population growth, economic development and industrialization, technological advancements, climate and environmental factors [2]. In South Africa, the leading contributor towards electricity generation is coal through the conversion of the chemical energy stored in coal. In the year 2024, coal accounted for approximately 82% of the country's electricity generation [3]. Coal has the highest percentage of energy generated. However, the constant reliance on coal brings challenges such as long-term sustainability, poses health risks, pollution and rising costs due to market demand.

Despite the high energy production, South Africa still faces an energy crisis driven by aging coal infrastructure, mismanagement at Eskom, who is responsible for over 80% of electricity generation, and insufficient generation capacity [1].

This is where renewable energy stands out. Renewable energy is energy that never runs out and can be used for a wide variety of applications, such as solar, wind and hydro.

To help meet this demand, engineers found a solution by using renewable energy. Renewable energy is energy derived from natural sources that are naturally replenishing, such as wind, solar and kinetic energy. Unlike solar and wind conditions which depend on the weather to generate electricity, human-powered energy generation will occur while people use the equipment. In this project, it is proposed to generate electrical energy from the kinetic energy produced by the oscillatory motion of a playground swing that will be low cost, low-resource, low maintenance, safe for kids and environmentally friendly. therefore, harvesting human-powered energy to produce electricity. This study investigates the design, simulation and implementation of the swing-based electrical power generator.

1.2 Objectives of this study

1.2.1 Problem Statement

The growing demand for electricity as well as the environmental and economic setbacks of fossil-fuel generation, has created a need for small-scale creative and innovative renewable energy solutions.

While large-scale solar and wind renewable energy is widely pursued, there still remains a significant gap in the development of low-cost energy harvesting systems that will be suitable for communities and educational applications. Kinetic energy produced during human activity in public spaces such as parks and playgrounds is part of a wide energy source that has been untouched. The pendulum-like motion that is generated by a person swinging contains mechanical energy which can be harvested and converted into electrical energy. This can serve as a practical and sustainable power source for low-power applications such as park lighting and phone charging station. Although the challenge lies in effectively converting the slow, bidirectional and irregular motion of a swing into smooth and usable electricity.

This study investigates the following key questions:

- How can the oscillatory kinetic energy of a swing be effectively converted to electrical energy?
- What mechanical design and configuration best optimises energy transfer from swing motion to the generator?
- Can a simulation model accurately predict the electrical energy generated from the system based on swing dynamics and operating conditions?
- Is the prototype capable of producing a sufficient electrical output for low-power application?

1.2.2 Purpose of the study

This study is significant for several connected reasons, such as energy access, environmental sustainability, engineering innovation and public education.

i. Energy access and community relevance

South Africa's national electricity access rate is approximately 84.4%, where rural areas have a much lower access rate of 75.3% [14]. This means that majority of the population, especially rural and underserved communities remain without reliable access to electricity. This is where this project of a proposed swing-based generator comes in, to serve as a small-scale energy harvesting system which offers as a practical and low-cost solution for supplying electricity to public spaces and off-grid communities that cannot be easily accessed by the conventional grid infrastructure.

ii. Environmental significance

The continue dominance of coal in South Africa's energy contributes significantly to greenhouse gases and air pollution, which brings about serious risks to public health and the environment. Energy harvesting devices can make use of renewable energy sources that are self-replenishing and abundant such as solar, wind, thermal and kinetic energy, which reduce the need for power transmission that cause energy losses and environmental impacts. Hy harnessing human kinetic energy, the swing generator being investigated in this study produces no harmful emissions and requires no fuel, making it an environmentally safe solution for electricity generation [15].

iii. Educational and social impact

Beyond the technical contribution, this study can be a major educational tool. Swings are enjoyed by people of all ages, making them an effective tool for demonstrating principles of energy conversion in a physical and engaging way. When the sing powers a light or a device in a park, it makes the concept of renewable energy visible for communities to observe and understand.

1.3 Scope and Limitations

Scope

This study focuses on the design, simulation and implementation of a small-scale swing-based electrical power generator that converts kinetic energy from the oscillatory motion of a playground swing into usable electrical energy. The scope of this study includes:

- The mechanical design of the energy conversion system.
- The electrical design of the output circuit.
- Development of the simulation model using Matlab/Simulink to predict the electrical output based on swing dynamics and operating conditions.
- Building a prototype of the swing generator.
- Evaluating the prototypes suitability for low-power applications such as powering a light.

Limitations

- Time constraints: The duration of this study is limited to one academic year which leaves less time for continuous testing and improvements.
- Budget constraints: The prototype is limited to low-cost components, which may not represent the best components that could be used in a real-life adaptation.
- Prototype scale: This study is for a small-scale lab model. Results may vary than a full-scale adaptation in a real playground.

2. Literature Review

2.1 Introduction to Renewable Energy Sources

Renewable energy is defined as the usable energy harvested from resources that are naturally replenishing such as solar, wind, hydro, geothermal heat, biomass [4]. Unlike fossil fuels such as coal, oil and natural gas, which accumulated after millions of years to form and are finite in supply, renewable energy sources are continuously replenished by nature and do not deplete over time [5]. An easily accessible source of renewable energy is a swing. Swings move in a pendulum-like motion, producing kinetic energy that can be transformed into electricity. Khan and Saeed et al [6] indicates that it is possible to access this free energy source without using fossil fuels by placing energy harvesting systems on swings in parks and playgrounds.

Mullane et al. [7] study states that implementing swing-based kinetic energy systems demonstrates the usefulness of renewable energy in a hands-on manner. According to [8], powering lighting and small electronics through swings reduces the area's burden on the traditional grid. Swings are enjoyed by people of all ages, which makes them a great tool for teaching people about the importance of saving energy and protecting the environment. When a swing is used to power lights or small devices in a park, it can help communities become more aware of environmental issues and encourage communities to think of new and innovative ways to generate energy.

2.2 Swing Power Generations

Swing power generation refers to the process of converting kinetic energy from the swinging motion of a swing and converting it into electrical energy [9]. This concept has gained attention through its high potential for generating sustainable electricity from human activities, in playgrounds and public spaces to be specific. The idea is to capture the kinetic energy produced by the oscillatory motion and convert it into electrical power for low-power applications.

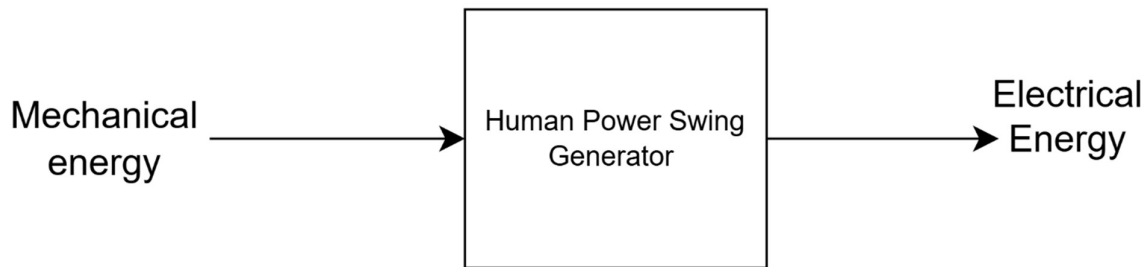


Figure 1: Simple block diagram showing conversion of energy

Rierner and Shapiro [10] explain the simple system behind power generation involves attaching a generator or other energy harvesting devices to the joints of the swing and as the swing moves back and forth, the motion causes the generator to spin and produce electricity. Swing power systems can make use of different types of swings to capture energy.

- Simple playground swings with hanging seats can generate power [11].

- More advanced pendulum-style swing that uses counterweights and fuller arm motions [11].

Some designs use dual generators at the top and bottom pivots of the swing to harness energy in both directions of motion. The study of swing motion by Shastri and Bharath [12], investigates two different designs to drive a mechanical and electrical load:

- Harnessing bi-directional motion of the swing.
- Harnessing any one direction of the swing.

Although, some engineering challenges include optimising the generator design for varying swing speeds as well as building a durable system to withstand weather conditions and years of use. Mortley [13] describes how to build a swing set that generates electricity by using an old swing set by hanging a drive shaft below the top of the swing and attaching swings to it by roller clutches. As the swing begins motion, it would turn the shaft forward, and when swung backwards, the roller clutches will keep motion, and the momentum of the generator keeps forward motion. As the motions continues back and forth, the generator generates electricity.

3. Methodology

3.1 Research Approach

This study follows an experimental engineering approach, along with theoretical design, computer-based simulation and practical prototype construction to investigate the effect of harvesting kinetic energy from a swing.

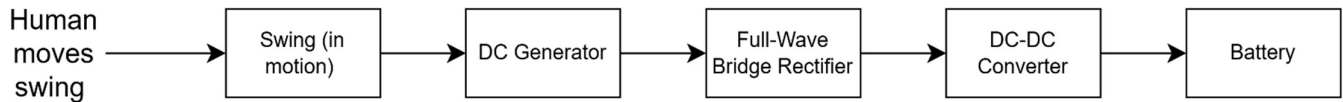


Figure 2: Block diagram showing the supposed system

Figure 3 shows the simplified block diagram of the proposed system for the swing power generator.

3.2 Specifications

Table 3-1: Comparison of Power supply:

Name:	Human Powered Swing:	Motor Driven swing:	Wind Powered swing:
Description:	A person sits on the swing and uses their body to make it move back and forth.	An electric motor is used to push the swing back and forth automatically.	Wind pushes the swing back and forth through a sail or paddle attached to the swing.
Advantage:	No fuel or electricity is needed. Anyone can use it and it teaches people about renewable energy in an interactive way.	The swing moves at a steady and controlled speed every time.	No person or motor is needed. It runs on its own as long as there is wind.
Disadvantages:	No power is produced when nobody is using the swing.	A power source is needed to run the motor, which means the system uses more electricity than it produces.	It only works when there is enough wind.
Why not chosen?	This method was chosen because it fits the criteria.	Using a motor to generate electricity would use more electricity than it produces, therefore defeating its purpose.	Wind is not always available.

Therefore, after considering different types of ways to cause the swing to move, it has been finalised to use a human-powered source to generate the kinetic energy in the swing.

Main components of this model:

- Swing frame
- Drive pulley (large)
- Generator pulley (small)
- Rubber belt
- DC generator
- Full wave bridge rectifier
- DC-DC converter (Buck converter)
- Rechargeable Battery

Table 3-2: Comparison of Swing Frame materials:

Name:	Metal frame:	Wooden Frame:	3D Printed Frame:
Description:	Made from aluminium that is welded or bolted together.	Made from wood that is joined together using screws or bolts.	Designed on a computer and printed using a 3D printer
Advantage:	Very strong and lasts a long time even under heavy use.	Simple to build using basic tools that is accessible.	Can be designed to exact measurements.
Cost:	Expensive.	Cheap.	Cheap.
Why not chosen:	Too heavy and expensive for a small-scale project. Welding is not always accessible.	Wood can absorb moisture and become misaligned.	This method was chosen.

Table 3-3: Comparison of different types of Drive pulleys (large):

Name:	Metal Pulley:	Wooden Pulley:	3D Printed Pulley:
Description:	Made from steel or aluminium.	Cut and shaped from wood.	Designed on a computer and 3D printed.
Advantage:	Strong and lasts a long time.	Cheap and easy to make with basic tools.	Made to exact size needed and easy to reprint if changes are needed.
Cost:	Expensive	Cheap	Cheap
Why not chosen:	Too expensive and needs special machining equipment.	Hard to get the exact shape and size right and warps over time.	This method is chosen.

Table 3-4: Comparison of different types of Generator Pulleys (small):

Name:	Metal Pulley:	Rubber Pulley:	3D Printed Pulley:
Description:	Made from steel or aluminium that attaches to the generator shaft.	Made from rubber that sits on the generator shaft	Designed on a computer and 3D printed
Advantage:	Very strong and does not wear down easily.	Grips the belt well and reduces slipping.	Can be printed to the exact size needed to achieve the correct speed ratio with the drive pulley.
Cost:	Expensive.	Cheap	Cheap
Why not chosen:	Too expensive and needs special equipment to machine to the correct small size.	Not rigid enough to maintain a consistent shape under repeated belt tension.	This method was chosen.

Table 3-5: Comparison of different types of DC-DC Converters:

Name:	Buck Converter:	Linear Voltage Regulator:	Boost Converter:
Description:	Steps voltage down to a stable lower level.	Reduces voltage to a fixed stable output level.	Steps voltage up to a higher level than the input.
Advantage:	Very efficient at converting higher voltages down.	Simple to use and very cheap with minimal components needed.	Useful when the generator output voltage is too low and needs to be increased.
Cost:	Cheap.	Very cheap.	Cheap.
Why not chosen:	This method is chosen.	Wastes excess voltage as heat making it inefficient for this application.	The generator is expected to produce sufficient voltage so stepping up is not needed.

Table 3-6: Comparison of DC Generator types:

Name:	Permanent Magnet DC Generator:	Stepper Motor Used Generator:	Brushless DC Generator:
Description:	Uses permanent magnets to produce electricity when the shaft is rotate.	Stepper motor that's repurposed to act as a generator by rotating its shaft to produce voltage.	Uses electronic commutation instead of brushes to produce electricity.
Advantage:	Simple to use.	Cheap and easy to find.	Very efficient and durable with no brushes to wear down.
Cost:	Low to moderate.	Very cheap.	Moderate to high.
Why not chosen:	This component was chosen.	Output voltage is irregular and difficult to control without additional processing circuits.	More expensive and requires additional electronic speed controllers.

3.2.1 Belt drive selection

For this project, an O- Ring belt was selected as most suitable for the belt and pulley system. Since the project is a small-scale prototype operating at low speeds and low loads, this type of belt drive is sufficient to transfer the rotational motion from the drive to the generator pulley. It is also very cheap and easy to replace if it stretches or breaks during testing, making it the most practical and cost effective choice for this prototype.

3.2.2 Full Wave Rectifier Selection

For this project a single package rectifier module was selected as the most suitable rectifier for the electrical output circuit. The rectifier module has four diodes assembled inside one component, making it easy to connect directly to the generator output and the battery charging circuit. Since the swing produces motion in both directions, the rectifier module converts both directions of current into a single direction of DC current suitable for charging the battery. The module is cheap, widely available and reduces the risk of loose connections that could occur during the vibration caused by the swing motion.

Table 3-7: Comparison of different types of Rechargeable Batteries:

Name:	Lead Acid Battery	Lithium Ion Battery	Nickel Metal Hydride Battery
Description:	Heavy rechargeable battery commonly used in cars	Lightweight rechargeable battery commonly used in phones	Rechargeable battery commonly used in household devices
Advantage:	Cheap and can handle being overcharged.	Lightweight and stores a large amount of energy.	Safe, reliable and widely available.
Cost:	Very cheap.	Expensive.	Moderate.
Why not chosen:	Too heavy and bulky for a small-scale prototype.	Too expensive for a small-scale prototype and requires a special charging circuit to prevent overcharging and damage.	This component was chosen.

3.3 Expected Design

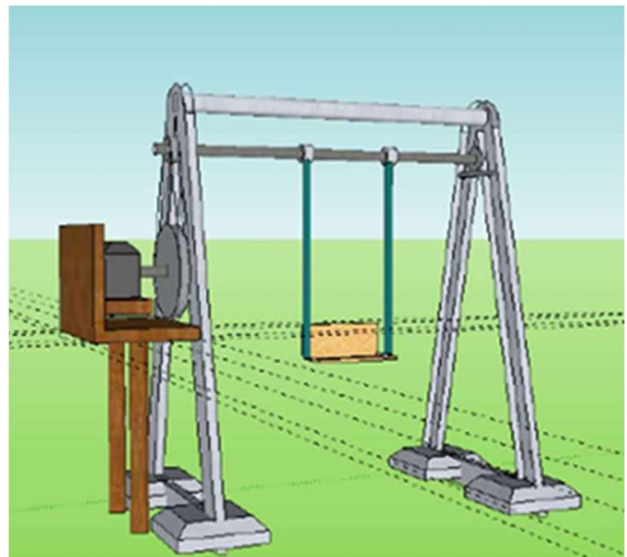
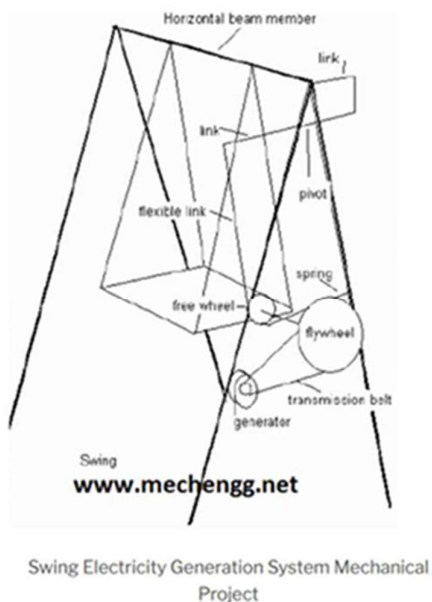


Figure 3: Mechanical layout of the proposed swing generator

4. Expected Results

It is anticipated that the proposed swing-based electrical power generator will successfully convert oscillatory motion into usable electrical energy through the designed swing power generator system. The model is expected to show that the oscillatory motion of the swing can be effectively transformed into rotational motion which drives the generator.

The system is expected to produce a measurable electrical output sufficient to power low-energy devices such as LED lights or small electronic components. It is further expected that the output power will vary depending on factors such as the amplitude and frequency of the swing motion, as well as the efficiency of the mechanical conversion mechanism.

Additionally, simulation and experimental results are expected to demonstrate the overall performance of the system.

5. Plan of Development

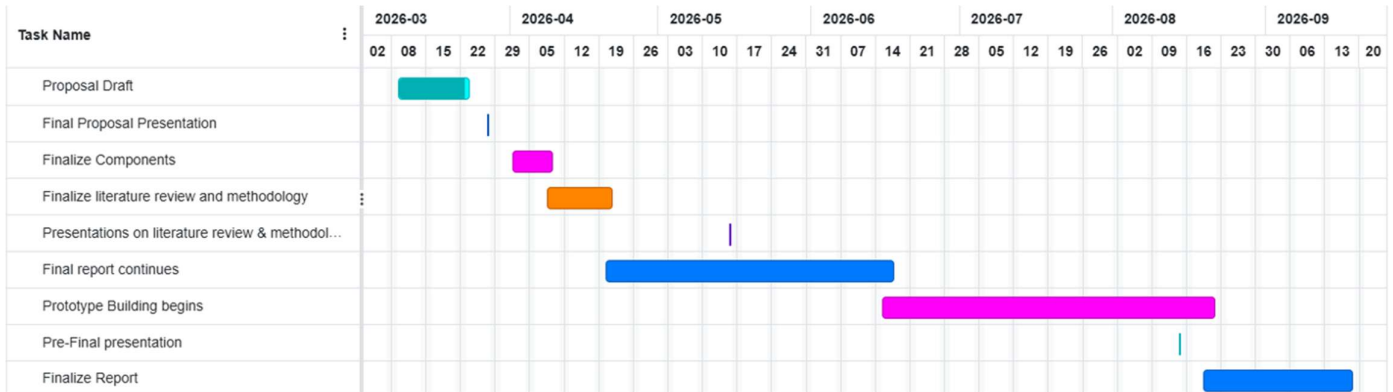


Figure 4: Gantt Chart of project

Summary of Gantt Chart:

- February, March: Proposal draft is written and proposal presentation.
- April, May: Finalising the literature review and methodology and finalise components as well as simulations. Presentation on literature review and methodology.
- April, May, June: Prototype building begins.
- September, October: Finalise report and pre-final presentation.
- October: Final project due.

6. Conclusion

In conclusion, this project proposes the development of a swing-based electrical power generator as an innovative approach to small-scale renewable energy harvesting applications. By converting human powered mechanical motion into electrical energy, the system aims to demonstrate a practical and sustainable energy solution. The proposed design includes kinetic energy conversion and electrical generation principles, supported by simulation and prototype testing. This project will aim to serve as an educational tool to help communities to understand renewable energy and how effective it is. It will also be a useful tool in powering low-power applications such as charging cell phones and powering lights.

7. List of References

- [1] "Understanding South Africa's Energy Crisis | African Energy Chamber," *energychamber.org*, Mar. 20, 2023. <https://energychamber.org/understanding-south-africas-energy-crisis/>
- [2] Internet Geography, "Why is energy consumption increasing?," *Internet Geography*, 2024. <https://www.internetgeography.net/topics/why-is-energy-consumption-increasing/>
- [3] "Utility-scale power generation statistics in South Africa," 2024. Available: https://www.csir.co.za/sites/default/files/2025-09/Utility%20Statistics%20Report_Jan%202025_Final.pdf
- [4] Encyclopaedia Britannica, "Renewable Energy," *Britannica.com*. [Online]. Available: <https://www.britannica.com/science/renewable-energy>. [Accessed: Mar. 2026].
- [5] H. Ritchie, M. Roser, and P. Rosado, "Renewable Energy," *Our World in Data*, 2020. [Online]. Available: <https://ourworldindata.org/renewable-energy>. [Accessed: Mar. 2026].
- [6] Khan, A., & Saeed, A. (2017). Self-energy sustainable playgrounds for children. *IEEE Xplore*. <https://ieeexplore.ieee.org/abstract/document/8324167/>.
- [7] Mullane, A., Bryans, G., & O'Malley, M. (2005). Kinetic energy and frequency response comparison for renewable generation systems. *IEEE Xplore*. <https://ieeexplore.ieee.org/abstract/document/1600525/>.
- [8] Emovon I. (2018) Comparative Performance and Analysis of Fuel Briquette Produced from Water Hyacinth Plant With Parafin Oil. *Journal of Advancement in Engineering and Technology*. Vol. 6., pp. 1-6.
- [9] Ijraset, I. (2020). Electricity generation by swing action. *International Journal for Research in Applied Science and Engineering Technology*, 8(6).
- [10] Riemer, R., & Shapiro, A. (2011). Biomechanical energy harvesting from human motion: Theory, state of the art, design guidelines, and future directions. *Journal of Neuroengineering and Rehabilitation*, 8, 22. <https://doi.org/10.1186/1743-0003-8-22>.
- [11] Bailey, A., Podlogar, M., Zarske, M., & Carlson, D. (2007). The science of swinging - Lesson. *TeachEngineering.Org*. https://www.teachengineering.org/lessons/view/cub_pend_lesson01
- [12] Shastri and Bharath, The Power of Swing, *International journal published under the patent corporation treaty*, 4645(292), 2009, 1-24.
- [13] Motley, Cleveland. 2011. "A Swing Set That Generates Electricity: 10 Steps (with Pictures)." *Instructables*. <https://www.instructables.com/A-Swing-Set-that-Generates-Electricity/>
- [14] Energy Capital Power, "The Ten Most Electrified Countries in Africa," *energycapitalpower.com*, Nov. 2023. [Online]. Available: <https://energycapitalpower.com/electricity-access-most-electrified-africa/>. [Accessed: Mar. 2026].
- [15] A. Ramella *et al.*, "The Power of Public Spaces: An Overview of Energy Production Potential," in *Springer*, Singapore, 2024. [Online]. Available